MILITARY SPECIFICATION

HOLDER ASSEMBLY, CHART-LET-DOWN, PLU-4/A AND PLU-5/A

1. SCOPE

1.1 Scope. This specification covers two types of book-type, chart-let-down holder assemblies.

1.2 Classification. The chart-let-down holder assemblies covered by this specification shall be of the following types, as specified (see 6.2):

PLU-4/A - Light provided by 28-volt lamps

PLU-5/A - Light provided by 6-volt lamps

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-576	Boxes, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-636	Box, Fiberboard
PPP-B-676	Boxes, Setup

Military

MIL-P-116	Preservation, Methods of
MIL-G-8602	Glass, Laminated, Flat, Aircraft
MIL-L-10547	Liners, Case, and Sheet, Overwrap: Water-vaporproof or Waterproof, Flexible
MIL-C-14806	Coating, Reflection Reducing, For Instrument Cover Glasses and Lighting Wedges
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MS3116	Connectors, Plug, Electric, Solder Type, Straight, Bayonet Coupling
MS25237	Lamp, Incandescent, Single Contact, Midget Flanged Base (T-1-3/4 Bulb)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Components. The holder assembly shall consist of a case, light, mounting attachment provisions, select switch, light-dimming rheostat and switch, filters, lens and lens frame, pencil holder, takeoff and landing data (TOLD) card holding provision, and electrical connections.

3.2 Materials. Materials used in the holder assembly shall be of uniform quality and condition and free of seams, cracks, or other defects which may adversely affect the appearance, strength, endurance, or wear resistance. Materials shall not be filled or otherwise treated to conceal defects nor shall they be composed of dissimilar metals as defined in MIL-STD-889.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Design and construction. The holder assembly shall be designed and constructed so that no parts will work loose in service. All parts subject to wear, breakage, or distortion shall be accessible to permit adjustment or replacement. The holder assembly shall be designed to hold a folded let-down chart securely in place so that no wrinkling, slipping, or vibrating of the chart will occur. The assembly shall also be totally operable with one hand, including stowing, unstowing, changing lamps, and book placement or removal. The book shall remain in place when the cover is opened and be easily removable by the user.

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3.4.1 Lighting system. The lighting system shall provide both red and white lighting, infinitely variable and independently controlled, and shall be such that no bright spots nor stray light are present. With the rheostat in the full-bright position, the brightest photometer reading shall not exceed 2.0 foot-lamberts and the dimmest photometer reading shall not be below 1.0 foot-lambert.

3.5 Performance

3.5.1 Dielectric strength. The holder assembly with lamps removed shall withstand a voltage of 556 volts, 60 cycles, applied between each connector pin and the holder assembly and between each pin for a period of 1 minute without evidence of leakage or breakdown.

3.5.2 Environmental conditions. The holder assembly shall be capable of satisfactorily performing its intended function and shall sustain no damage or detrimental effect when subjected to the following environmental conditions:

a. Vibration incident to service usage

b. Temperatures ranging from -55°C to +70°C

c. Rainfall as encountered in any locale

d. Exposure to warm, highly humid atmospheres such as encountered in tropical areas

e. Exposure to fungus growth as encountered in tropical climates

f. Exposure to salt-fog atmosphere

g. Dust particles as encountered in desert areas.

3.5.2.1 Explosive atmosphere. The holder assembly shall be capable of operating in the presence of an explosive atmosphere without creating an explosion and shall be capable of containing an explosion occurring inside the holder assembly.

3.6 Details of components. The holder assembly shall be furnished as a complete unit. Any components required for operation of the unit shall be included even though they may not be listed herein.

3.6.1 Case. The case shall be constructed from a nonmagnetic material and shall serve as a frame for the lights, rheostat, select switch, electrical

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input, TOLD card, and lens. The case shall have mounting attachment provisions so that it may be used for either the pilot or copilot station (see figure 1).

3.6.2 Light. Light shall be provided by multiple MS25237-327-15, 28-volt lamps for the PLU-4/A holder assembly and by multiple MS25237-328-AS15, 6-volt lamps for the PLU-5/A assembly. The lamps shall be replaceable by an operator wearing gloves.

3.6.3 Mounting attachment provision. Mounting attachment provisions shall be provided on both sides and center back of the holder assembly and shall be designed to permit easy installation and removal of the mounting attachment bracket (see figure 1).

3.6.4 Select switch. The select switch shall provide selection of either red or white light and shall be operable by an operator wearing gloves.

3.6.5 Rheostat and switch. A rheostat, using its full mechanical range, shall be provided to control the brightness of the light. The rheostat shall have a built-in switch for cutting the light off and shall be operable by an operator wearing gloves. The rheostat shall operate smoothly and uniformly throughout its range of travel both electrically and mechanically. Both red and white lighting shall be fully dimmable to the off position of the rheostat.

3.6.6 Filters. A red filter shall provide aviation red light as specified in MIL-L-25467. A white filter shall provide instrument white light as specified in MIL-L-27160.

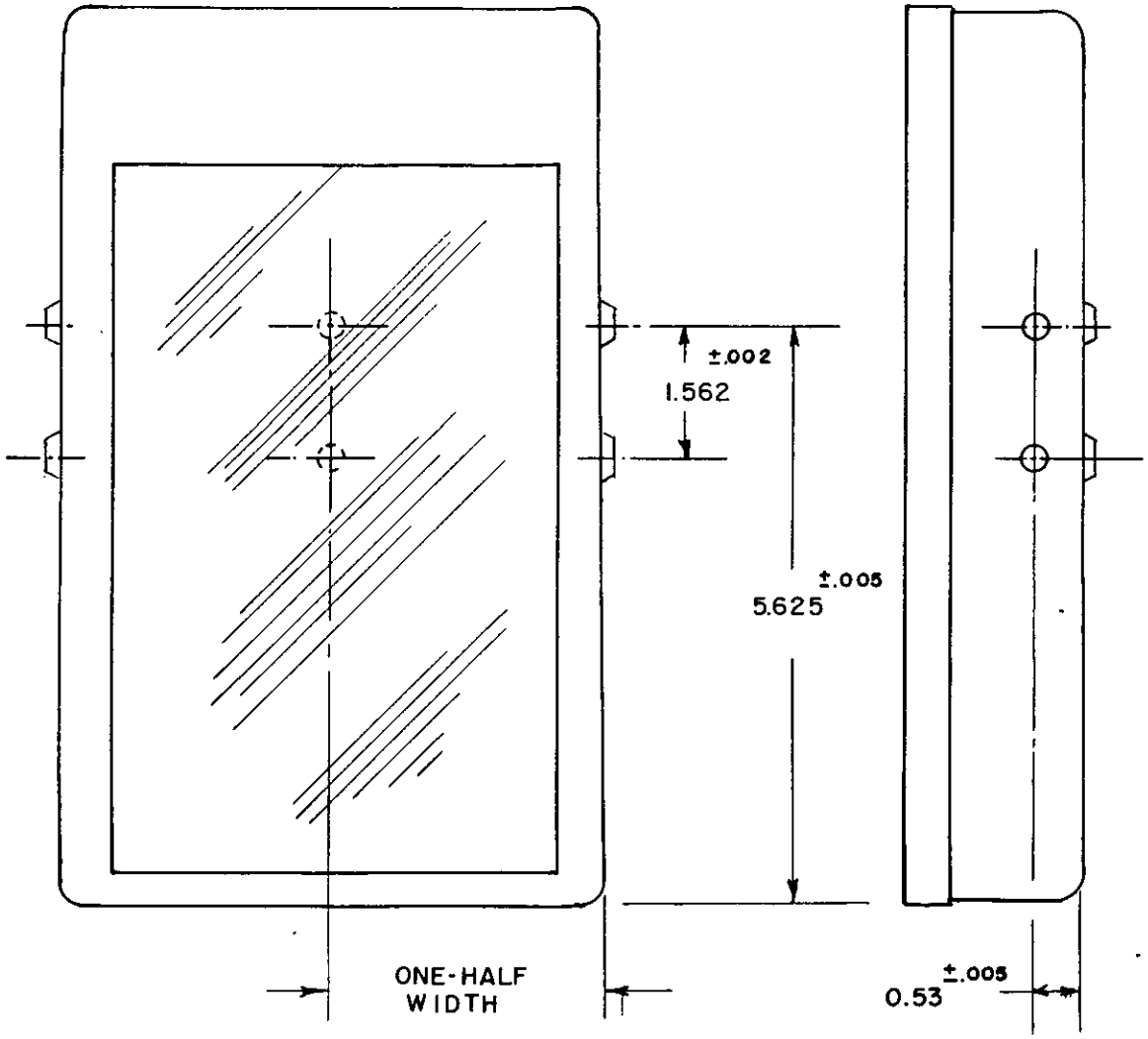
3.6.7 Lens. The lens shall be made of safety glass in accordance with MIL-G-8602 and shall be mounted so that there is no movement within the frame. The lens shall be coated in accordance with MIL-C-14806.

3.6.8 Electrical connections. The holder assembly shall be provided with a 2-conductor, No. 18 AWG, 7-ampere, 300-volt maximum, rubber-jacketed, retractile power cord with an MS3116E8-3P plug. The cord shall be 3 feet in length when fully extended.

3.6.9 Pencil holder. A pencil holder shall be provided to hold pencils of varying diameters regardless of the holder assembly attitude.

3.6.10 TOLD card provision. TOLD card provisions shall be incorporated to hold a clear plastic TOLD card, measuring 5 inches by 8 inches by 1/16 inch, over the cover lens so that no movement is incurred while writing and shall permit easy installation and removal.

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NOTE: SIX HOLES, TAP .190-32 UNF-3B

3/16 MIN THREAD DEPTH

DIMENSIONS IN INCHES

FIGURE 1. Chart-Let-Down Holder Mounting Provisions

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3.7 Dimensions. The overall dimensions, excluding mounting attachment provision, shall not exceed 10 inches high by 5.85 inches wide by 1.9 inches thick. The dimensions of the viewing area shall be a minimum of 5 inches wide by 8 inches high.

3.8 Weight. The weight of the holder assembly shall not exceed 2.5 pounds.

3.9 Color. The color of the case shall be lusterless black color No. 37038 in accordance with FED-STD-595.

3.10 Identification of product. Assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.11 Workmanship. Standards of workmanship shall be such that the holder assemblies will comply with the requirements and tests specified herein. Particular attention shall be given to freedom from defects in materials and parts; freedom from binding, burrs and sharp edges, defective connections, and loose, damaged, or missing parts; and adequacy of finishes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the holder assembly shall be classified as quality conformance tests.

4.3 Quality conformance tests. Quality conformance tests shall consist of the following:

a. Individual tests

b. Sampling tests.

4.3.1 Individual tests. Each holder assembly shall be subjected to the examination of product test as described in 4.4.1.

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4.3.2 Sampling tests. Holder assemblies shall be selected at random from each inspection lot as specified in 4.3.2.1 and subjected to the following tests as described under 4.4:

a. Examination of product

b. Lighting test

c. Environmental tests.

4.3.2.1 Lot size and number of test samples. The lot size and number of samples to be selected for sampling tests shall be as follows:

Lot Size	Number of Samples
25 or less	1
26 to 50	2
51 to 100	3
100 or more	3 plus 1 for each 100 or fraction thereof

4.3.2.2 Rejection and retest. When one or more items from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld until the extent and cause of failure have been determined. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After correction all sampling tests shall be repeated.

4.3.2.3 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot or lots produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.3.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.4 Test methods

4.4.1 Examination of product. Each holder assembly submitted for acceptance shall be examined for the following defects. If the holder assembly has one or more defects, it shall be rejected.

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a. Component missing

b. Light does not illuminate

c. Light does not dim properly

d. Cracked or broken lens

e. Cracked or broken case

f. Materials not as specified

g. Workmanship not acceptable.

4.4.2 Lighting test

4.4.2.1 Dielectric strength. Each lamp shall be removed from its socket and a test voltage of 556 volts, 60 cycles, connected between each connector pin and the holder assembly and between each pin for a period of 1 minute. There shall be no evidence of leakage or breakdown as a result of this test.

4.4.2.2 Operating controls. The operating controls shall be subjected to the following tests:

a. The on/off rheostat shall be operated to determine that it performs smoothly and uniformly throughout its range of travel both electrically and mechanically.

b. Both red and white lighting shall be checked to determine that both types of lighting are fully dimmable to the off position of the rheostat.

c. The color selector switch shall be operated to determine that it will select red corresponding with red illumination and white corresponding with white illumination.

4.4.2.3 Brightness measurements. The following tests shall be performed for both red and white lighting:

a. Insert a 5- by 7-inch piece of white calibration paper, reflectance of 85 ± 5 percent, comparable to FED-STD-595 graduated into 1- by 1-inch squares, into the holder assembly.

b. Depending on the lamp used, apply an operating voltage of 28 ±.5 volts or 6 ±.05 volts to the mating connector of the holder assembly.

c. Position the photometer and individually focus on the approximate center of each graduated square of the calibrating paper. Rotate the brightness

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control rheostat to the full-bright position. Measure and record the brightness at each point. The brightest photometer reading shall not exceed 2.0 foot-lamberts. The dimmest photometer reading shall not be below 1.0 foot-lambert.

4.4.2.4 Stray light. Stray light measurements shall be in accordance with the requirement entitled "Stray light" of MIL-L-25467, with lamps operating at rated voltage.

4.4.3 Environmental tests. Unless specific test methods of MIL-STD-810 are specified, environmental tests shall be in accordance with the general requirements of section 3 of MIL-STD-810.

4.4.3.1 Vibration. A booklet of charts shall be inserted into the holder assembly. The entire assembly shall be mounted by the mounting attachment to a vibration exciter and subjected to vibration in accordance with the following schedule:

Frequency Range	Acceleration
5-2,000 Hz	5g

4.4.3.1.1 Resonance search. The vibration exciter shall be programmed to cycle slowly through the range of specified vibration. The holder assembly (operating, lights on) shall be visually monitored to detect any resonance. When resonances are detected, the actual frequency shall be recorded.

4.4.3.1.2 Resonance dwell. The holder assembly shall be vibrated for a period of not less than 1 hour at each resonant frequency. If no resonances are encountered, the assembly shall be subjected to not less than 3 hours of cyclic vibration from 5 through 200 Hz at 5g. The time required for 1 cycle shall not exceed 10 minutes.

4.4.3.1.3 Observations. During the vibration period, there shall be no evidence of intermittent lighting or erratic control functions. After the vibration period, the assembly holder shall be thoroughly examined for evidence of deterioration or damage. Failure to comply with any of these requirements shall constitute a failure and shall be so recorded.

4.4.3.2 High temperature. The holder assembly shall be mounted within an environmental chamber and the internal temperature of the chamber raised to $70^{\circ}C \pm 2^{\circ}C$ for a period of not less than 4 hours. At the completion of the exposure period, the viewing area of the holder assembly shall be visually observed. There shall be no evidence of fogging or mist on the lens.

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4.4.3.2.1 The holder assembly shall be allowed to return to room temperature and then thoroughly examined. There shall be no evidence of deterioration or damage as a result of the high temperature test.

4.4.3.3 Low temperature. The holder assembly (not operating) shall be mounted within an environmental chamber and the internal temperature of the chamber lowered to -55°C ±2°C for a period of not less than 4 hours. At the completion of the exposure period, the viewing area of the holder assembly shall be visually observed. There shall be no evidence of fogging or mist on the lens. While still at -55°C, the lights shall be turned on for 5 minutes and the assembly re-examined; there shall be no evidence of fogging or mist on the lens.

4.4.3.3.1 The holder assembly shall be allowed to return to room temperature and then thoroughly examined. There shall be no evidence of deterioration or damage as a result of the low temperature test.

4.4.3.4 Explosive atmosphere. The holder assembly shall be subjected to an explosive atmosphere test in accordance with method 511, procedure I of MIL-STD-810.

4.4.3.5 Rain. The holder assembly shall be subjected to a rain test in accordance with method 506 of MIL-STD-810.

4.4.3.6 Humidity. The holder assembly shall be subjected to a humidity test in accordance with method 507, procedure I of MIL-STD-810.

4.4.3.7 Fungus. The holder assembly shall be subjected to a fungus test in accordance with method 508 of MIL-STD-810.

4.4.3.8 Salt fog. The holder assembly shall be subjected to a salt fog test in accordance with method 509 of MIL-STD-810.

4.4.3.9 Dust. The holder assembly shall be subjected to a dust test in accordance with method 510 of MIL-STD-810.

4.5 Inspection of preservation, packaging, packing, and marking. Sample items or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

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5.1.1 Level A. Each holder assembly shall be packaged method IC1 in accordance with MIL-P-116 (without preservative compound) and overboxed in a snug-fitting unit container conforming to PPP-B-676.

5.1.2 Level C. The holder assemblies shall be packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice provided the latter meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Holder assemblies packaged as specified in 5.1.1 shall be packed in exterior overseas-type shipping containers conforming to PPP-B-576 or PPP-B-636. Insofar as practicable, exterior shipping containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required, and shall contain identical quantities. The gross weight of each shipping container shall not exceed the limitation of the container specification. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers conforming to PPP-B-576 shall be provided with a case liner conforming to MIL-L-10547 and sealed in accordance with the appendix thereto.

5.2.2 Level B. Holder assemblies packaged as specified in 5.1.1 shall be packed in domestic-type exterior shipping containers conforming to PPP-B-576 or PPP-B-636. Exterior shipping containers shall be of minimum cube and tare consistent with the protection required. Insofar as practicable, exterior shipping containers shall be of uniform shape and size and shall contain identical quantities. The gross weight of each shipping container shall not exceed the limitation of the container specification. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. When fiberboard containers are used, such containers shall meet the special requirements table of PPP-B-636, as applicable.

5.2.3 Level C. Holder assemblies shall be packed in a manner that will afford adequate protection at the lowest rate against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice provided the latter meets the requirements of this level.

5.3 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The holder assemblies covered by this specification are intended to hold documents for visual reference by pilots and copilots during flight in various aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Type of holder assembly (see 1.2)

c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

Custodian:
Air Force - 11

Review activity:
Air Force - 82

Preparing activity:
Air Force - 11

Project No. 1680-F217

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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